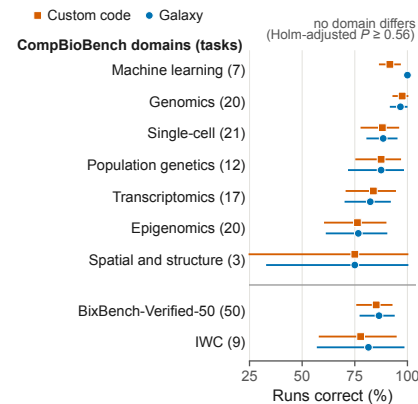
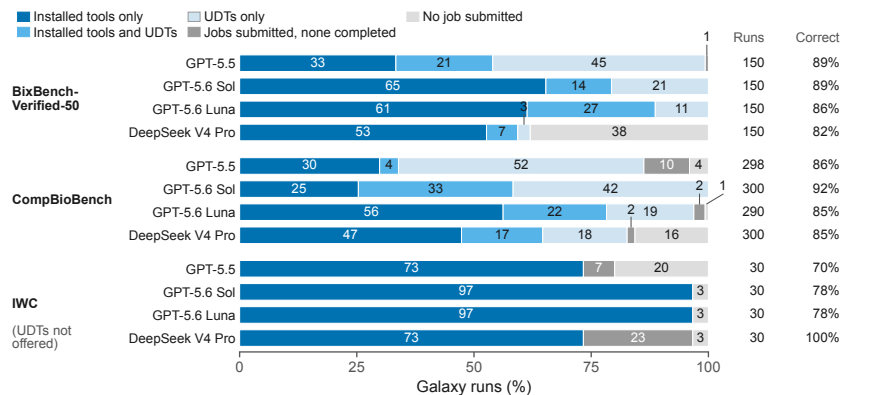


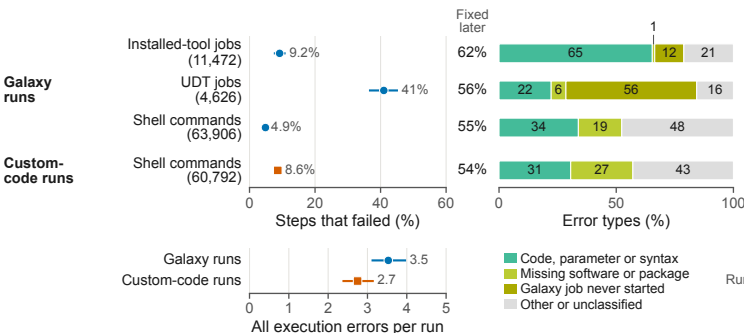
a Correct runs by domain



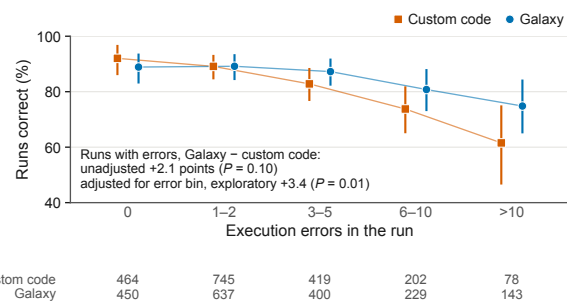
b How each Galaxy run used Galaxy, and how it ended



c How often execution steps failed, and were fixed later

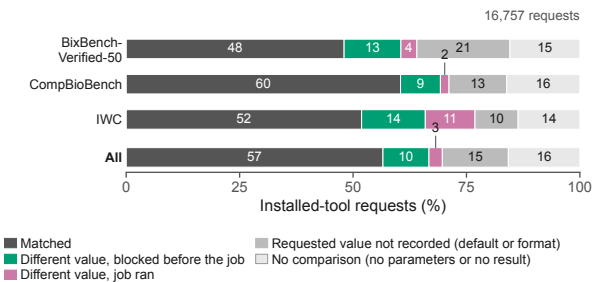


d Final correctness among runs with execution errors



e Parameter checks on installed-tool requests

The interface compares the requested parameters with those Galaxy validated or recorded



After a request was blocked for a different value, the run later completed a job of the same tool in 71% of 1,700 cases (with matching parameters in 52%). Matching parameters do not show that the settings were scientifically appropriate.

f Failed requests and candidate infrastructure improvements

Failure classes grouped by the change most likely to help (codebook in Source Data); an independent rater reproduced 92% of the classes and named the same improvement for 71% of requests

